



Florida Clinicians for Climate Action

Florida Clinicians for Climate Action's Medical Student Community Based Climate and Health Practicum

The 2022 AAMC survey on medical schools' response to climate change showed only 55% of US medical schools included instruction on its health impacts. Although this number reflects a near doubling of medical school responses in five years, it is far lower than expected, for a problem that threatens human existence. Even more obviously absent from the climate and health curricula is a community based experiential opportunity. Medical students have reported feeling ill-prepared to manage climate-related health concerns in patient care and have been the catalyst for curricula improvement.

Recognizing the need, the Global Consortium on Climate and Health Education (GCCHE) — launched in 2017 by Columbia University Mailman School of Public Health following the 2015 COP-21 conference in Paris — developed a comprehensive framework of core competencies to guide health professions education. Now in its fourth iteration and representing over 240 member institutions in 50+ countries, the GCCHE competency framework encompasses five essential domains — Knowledge and Analytic Skills, Collaboration and Communication, Policy, Public Health Practice, and Clinical Practice — providing scientific rationale, shared language, and evidence-based knowledge needed to advance both adaptation and mitigation in healthcare.

Against this backdrop of documented educational gaps, in 2023, the Florida Clinicians for Climate Action (FCCA) developed the Project C-HOT (Climate–Heat Outreach Team) Medical Student Practicum, a volunteer capstone community health outreach opportunity grounded in the GCCHE Core Competency Framework. This practicum gives students the opportunity to apply climate and health knowledge in real-world, community-facing settings under structured, competency-based supervision. This voluntary capstone opportunity allows students to become active agents of community health improvement and climate action.

The practicum offers value on each of the five domains:

- **Knowledge and Analytic Skills**— Medical students complete three hours of foundational knowledge of climate change, especially heat-related illnesses upon entry into the practicum. Throughout the year long experience they receive opportunities to participate in ongoing educational programs for Op-Ed writing, community-level GIS mapping and resilience planning, and integration of climate-health information into patient assessment and care plans.
- **Collaboration and Communication**— South Florida's communities face disproportionate vulnerability to extreme heat and climate-related health threats. The practicum places future physicians directly in these communities, alongside other FCCA's clinicians (doctors, nurses, pharmacists, public health professionals) to teach about climate change health impacts, increasing health literacy, connecting vulnerable populations to protective resources, and building local resilience capacity.
- **Policy**— FCCA encourages all clinicians to advocate for policies that help eliminate the factors underlying planetary warming while ensuring speedy implementation of equitable solutions that improve health outcomes for those already suffering from the change. Consequently, the students may choose to assist in existing

campaigns. The practicum builds the pipeline of climate-informed clinicians that the literature identifies as urgently needed. The 2024 scoping review in the Journal of Graduate Medical Education found a striking absence of rigorous competency-based GME climate curricula, making early formative experiences such as this practicum especially critical.

- **Public Health Practice**— The practicum model, curriculum, and outcomes are designed to generate peer reviewed scholarship, closing the research gap identified in published literature regarding the efficacy of community-integrated climate health education for medical students.
- **Clinical Practice**- FCCA partners with free clinics and federally qualified health centers to provide educational activities on climate and health for their patients. The students can participate at least one event per week of varying activities from webinars, lobby teaching, health fairs to community organizations' presentations.

Program Outcomes

Herbert Wertheim College of Medicine's student interest group Physician for Social Responsibility obtained approval for FCCA's medical student practicum and yearly five to seven students participate. The FCCA Project C-HOT Practicum was designed from the outset to generate outcomes that extend beyond individual student learning to contribute to the broader scientific and public discourse on climate change and health. The practicum's outcomes span three interconnected domains:

1. Research Activities and Peer-Reviewed Publications

A core feature of the practicum curriculum is the integration of research activity with community health outreach. Students participated in structured data collection and analysis activities grounded in the Public Health Practice domain of the GCCHE framework, including use of GIS mapping to identify heat-vulnerable communities and engagement with FCCA's ongoing community resilience research. These research activities were designed to support peer-reviewed publication, providing students with scholarly authorship experience while advancing the evidence based on community-level climate health interventions in South Florida. Two peer reviewed articles

1. **Heat Illness and Extreme Weather Health Literacy: Communication Preferences and Effectiveness for Patients Living in Climate-Change-Vulnerable Communities** *Int. J. Environ. Res. Public Health* 2025, 22, 434
2. **Jaramillo Arias M, Kulkarni N, Le A, et al. (April 03, 2024) Climate Change, Emerging Vector-Borne Illnesses, and Anesthetic Considerations. Cureus 16(4): e57517. doi:10.7759/cureus.57517**

2. Op-Eds and Policy Engagement

Offering students a partnership with experienced clinicians to write Op-eds or letter to the editor (after a didactic) directly operationalized in the practicum. Yearly, the student co-authored Op-eds on a health impact were published in local and national news outlets. Through hands-on experience, they understood the call for trusted voices of health professionals to engage urgently in policy advocacy.

1. [Bridging the gap between climate education and maternal care | The Invading Sea](#)
2. <https://www.theinvadingsea.com/2024/03/26/whale-pump-conveyor-belt-climate-change-carbon-emissions-phytoplankton-health/>

3. <https://www.theinvadingsea.com/2025/08/19/doulas-pregnancy-heat-extreme-weather-emergency-plans-florida-clinicians-for-climate-action/>

3. Media Articles and Public Communication

The Knowledge and Analytical Skills domain of the practicum curriculum required students to demonstrate their ability to communicate climate-health information in concise, accessible formats for social media and mass media audiences. Students produced short-form videos and media-ready content explaining the health impacts of climate change using the HEATWAVE mnemonic framework. Media articles and coverage resulting from practicum activities served to amplify community awareness, extend the reach of FCCA's mission, and provide students with real-world science communication experience.

The practicum model offers the student experiences with helping communities that are impacted now by climate change while improving their climate literacy, closing research gaps, and strengthening their analytical skills and achieving important competencies that are essential for 21st-century physicians.

References:

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3. Greenwald L, Blanchard O, Hayden C, Sheffield P. Climate and health education: A critical review at one medical school. *Front Public Health*. 2023;10:1092359. doi: 10.3389/fpubh.2022.1092359. 4. Global Consortium on Climate and Health Education (GCCHE). Core Concepts Framework. Columbia University Mailman School of Public Health; 2022 (4th iteration). Available at: <https://www.publichealth.columbia.edu/research/programs/global-consortium-climate-health-education/resources/core-concepts>.
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6. Wellbery C, Sheffield P, Timmireddy K, Sarfaty M, Teherani A, Fallar R. *It's Time for Medical Schools to Introduce Climate Change Into Their Curricula*. *Academic Medicine* 93(12):1774-1777, December 2018. DOI: 10.1097/ACM.0000000000002368.